

PSEUDORANDOM NUMBERS.Formula

$$x_{n+1} = (ax_n + c) \bmod m$$

let $x_0 = 3$, $a = 7$, $c = 4$, $m = 9$

Solution:when $n = 0$

$$x_{0+1} = (ax_0 + c) \bmod m$$

$$x_1 = (7 \times 3 + 4) \bmod 9 = 25 \bmod 9 = 7$$

$$x_1 = 7$$

when $n = 1$

$$x_{1+1} = (ax_1 + c) \bmod m$$

$$x_2 = (7 \times 7 + 4) \bmod 9 = 53 \bmod 9 = 8$$

$$x_2 = 8$$

when $n = 2$

$$x_{2+1} = (ax_2 + c) \bmod m$$

$$x_3 = (7 \times 8 + 4) \bmod 9 = 60 \bmod 9 = 6$$

$$x_3 = 6$$

when $n = 3$

$$x_{3+1} = (ax_3 + c) \bmod m$$

$$x_4 = (7 \times 6 + 4) \bmod 9 = 46 \bmod 9 = 1$$

$$x_4 = 1$$

when $n = 4$

$$x_{4+1} = (ax_4 + c) \bmod m$$

$$x_5 = (7 \times 1 + 4) \bmod 9 = 11 \bmod 9 = 2$$

$$x_5 = 2$$

So on ...

HASHING FUNCTION

Formula

$$h(K) = K \bmod m$$

m = memory locations , K = Keys (values)

$K = 42, 53, 65, 21, 30, 69, 56, 36, 28, 44$
 $\Rightarrow \boxed{m = 10}$

$$h(42) = 42 \bmod 10 = 2$$

$$h(53) = 53 \bmod 10 = 3$$

$$h(65) = 65 \bmod 10 = 5$$

$$h(21) = 21 \bmod 10 = 1$$

$$h(30) = 30 \bmod 10 = 0$$

$$h(69) = 69 \bmod 10 = 9$$

$$h(56) = 56 \bmod 10 = 6$$

$$h(36) = 36 \bmod 10 = 6$$

$$h(28) = 28 \bmod 10 = 8$$

$$h(44) = 44 \bmod 10 = 4$$

$$6 + 1 = \textcircled{7}$$

By Applying linear Probing

$$h(K, i) = (h(K) + i) \bmod m$$

Keys	30	21	42	53	44	65	56	36	28	69
m	0	1	2	3	4	5	6	7	8	9

Encryption Function:

$$f(P) = (P + K) \bmod 26$$

when $K = 3$

$$f(P) = (P + 3) \bmod 26$$

"UNIVERSITY" Encrypted Message

For U; $f(20) = (20 + 3) \bmod 26 = 23 \bmod 26 = 23$ (X)
For N; $f(13) = (13 + 3) \bmod 26 = 16 \bmod 26 = 16$ (Q)
For I; $f(8) = (8 + 3) \bmod 26 = 11 \bmod 26 = 11$ (L)
For V; $f(21) = (21 + 3) \bmod 26 = 24 \bmod 26 = 24$ (Y)
For E; $f(4) = (4 + 3) \bmod 26 = 7 \bmod 26 = 7$ (H)
For R; $f(17) = (17 + 3) \bmod 26 = 20 \bmod 26 = 20$ (U)
For S; $f(18) = (18 + 3) \bmod 26 = 21 \bmod 26 = 21$ (V)
For I; $f(8) = (8 + 3) \bmod 26 = 11 \bmod 26 = 11$ (L)
For T; $f(19) = (19 + 3) \bmod 26 = 22 \bmod 26 = 22$ (W)
For Y; $f(24) = (24 + 3) \bmod 26 = 27 \bmod 26 = 1$ (B)

Decrypted Message: "XQLYHUVLWB"

DECRYPTION FUNCTION:

$$F^{-1}(P) = (P - K) \bmod 26$$

when $K = 5$

$$F^{-1}(P) = (P - 5) \bmod 26$$

Decrypted Message ~~to be~~ "JCFR"

For J; $f(9) = (9 - 5) \bmod 26 = 4 \bmod 26 = 4$ (E)

For C; $f(2) = (2 - 5) \bmod 26 = -3 \bmod 26 = 23$ (X)

For F; $f(5) = (5 - 5) \bmod 26 = 0 \bmod 26 = 0$ (A)

For R; $f(17) = (17 - 5) \bmod 26 = 12 \bmod 26 = 12$ (M)

Encrypted Message "EXAM"

G.C.D "Greatest Common Divisor" (HCF)

$$24 = 2 \times 2 \times 2 \times 3$$

$$36 = 2 \times 2 \times 3 \times 3$$

$$\text{GCD} = 2 \times 2 \times 3 = 12.$$

$$17 = 1 \times 17$$

$$22 = 2 \times 11$$

$$\boxed{\text{GCD} = 1}$$

L.C.M -

$$24 = 2 \times 2 \times 2 \times 3$$

$$36 = 2 \times 2 \times 3 \times 3$$

$$\text{L.C.M} = \text{Common} \times \text{Uncommon}$$

$$= (2 \times 2 \times 3) \times (2 \times 3) = 12 \times 6 = 72.$$

$$\text{G.C.D. of } (a, b) = P_1^{\min(a_1, b_1)} P_2^{\min(a_2, b_2)} \dots P_n^{\min(a_n, b_n)}$$

$$\text{L.C.M. of } (a, b) = P_1^{\max(a_1, b_1)} P_2^{\max(a_2, b_2)} \dots P_n^{\max(a_n, b_n)}$$

$$120: 2 \times 2 \times 2 \times 3 \times 5 = 2^3 \cdot 3 \cdot 5$$

$$500: 2 \times 2 \times 5 \times 5 \times 5 = 2^2 \cdot 5^3 \cdot 3^0 \quad \therefore 3^0 = 1$$

$$\text{Gcd}(120, 500) = 2^{\min(3, 2)} \cdot 3^{\min(1, 0)} \cdot 5^{\min(1, 3)}$$

$$= 2^2 \cdot 3^0 \cdot 5^1 = 4 \times 1 \times 5 = 20.$$

$$\text{Lcm}(120, 500) = 2^{\max(3, 2)} \cdot 3^{\max(1, 0)} \cdot 5^{\max(1, 3)}$$

$$= 2^3 \cdot 3^1 \cdot 5^3 = 8 \times 3 \times 125 = 3,000.$$

$$\text{Gcd}(95256, 432) = 2^{\min(3, 4)} \cdot 3^{\min(5, 3)} \cdot 7^{\min(2, 0)}$$

$$(2^3 3^5 7^2, 2^4 3^3) = 2^3 \cdot 3^3 \cdot 7^0 = 8 \times 27 \times 1 = 216.$$

$$\text{L.C.M.}(95256, 432) = 2^{\max(3, 4)} \cdot 3^{\max(5, 3)} \cdot 7^{\max(2, 0)}$$

$$(2^3 3^5 7^2, 2^4 3^3) = 2^4 \cdot 3^5 \cdot 7^2 = 16 \times 243 \times 49 = 190512.$$

Euclidean Algorithm

compute. $\text{Gcd}(120, 500)$:

$\text{Gcd}(a, b)$

- $\therefore$ Greater number will be dividend,
- $\therefore$ Smaller number will be divisor.

$$a = qd + r$$

Division Algorithm

$$500 = () (120) + ()$$

$$500 = (4) (120) + (20)$$

$$120 = () (20) + ()$$

$$120 = (6) (20) + (0)$$

↓
stop

$$\text{Gcd}(120, 500) = 20 \quad \text{Ans}$$

x — x — x — x — x — x —

compute.

$\text{Gcd}(91, 287)$

$$a = qd + r$$

$$287 = (3) (91) + 14$$

$$91 = (6) (14) + (7)$$

$$14 = (2) (7) + 0$$

↓
stop

$$\text{Gcd}(91, 287) = 7 \quad \text{Ans}$$

Gcd as linear combinations

Bezout Theorem

$$\gcd(a, b) = \overline{S}a + \overline{t}b$$

~~Find~~ Gcd(6, 14) as linear combination: Bezout coefficients

First apply Euclidean algorithm to calculate the gcd(a, b)

$$a = qd + r$$

$$14 = (2)(6) + 2$$

$$6 = (3)(2) + 0$$

$$\gcd(6, 14) = 2$$

Now rewrite equation in terms of r

$$2 = (1)(14) - (2)(6)$$

$$2 = \underbrace{1 \cdot 14 + (-2) \cdot 6}_{\text{Bezout coefficient}}$$

x — x — x — x — x — x — x — x — x

Show $\gcd(252, 198) = 18$ as linear combination of 252 and 198

Sol

$$a = qd + r$$

$$252 = (1)(198) + 54$$

$$198 = (3)(54) + 36$$

$$54 = (1)(36) + 18$$

$$36 = (2)(18) + 0$$

Now rewrite equation in terms of r

$$54 = 1 \cdot 252 - 1 \cdot 198 \quad \text{--- iii}$$

$$36 = 1 \cdot 198 - 3 \cdot 54 \quad \text{--- ii}$$

$$18 = 1 \cdot 54 - 1 \cdot 36 \quad \text{--- i}$$

Now substitute eqn (ii) & (iii) in (i)

$$18 = 1 \cdot 54 - 1 \cdot 36 = 1 \cdot 54 - 1 \cdot (1 \cdot 198 - 3 \cdot 54)$$

$$18 = 1 \cdot 54 - 1 \cdot 198 + 3 \cdot 54 = 4 \cdot 54 - 1 \cdot 198$$

$$18 = 4(1 \cdot 252 - 1 \cdot 198) - 1 \cdot 198 = 4 \cdot 252 - 4 \cdot 198 - 1 \cdot 198$$

$$18 = 4 \cdot 252 - 5 \cdot 198$$

$$\boxed{18 = (4)(252) + (-5)(198)} \quad \text{Ans}$$

1 Linear Congruencies

$$ax \equiv b \pmod{m}$$

Find x .

Solve for x : $3x \equiv 4 \pmod{7}$.

Sol: First we have find inverse of a .

Here, $a=3$, $b=4$, $m=7$

To compute first check that $\gcd(a, m) = 1$.

$$\gcd(a, m) = \gcd(3, 7) = 1$$

$$a = qd + r$$

$$7 = (2)(3) + \textcircled{1} \rightarrow \text{Hence inverse exists.}$$

Now we can calculate inverse like,

$$\gcd(a, m) = 1 = as + tm$$

Now $1 = 1 \cdot 7 - 2 \cdot 3$

$$1 = (1)(7) + (-2)(3)$$

~~$1 = 1 \cdot 7 - 2 \cdot 3$~~
 $\gcd(3, 7) = 1 = t m + s a$

inverse can be make positive
by adding mod value.

$$\Rightarrow \bar{a} = -2, m=7$$

$$\text{Hence } \bar{a} = -2 + 7 = 5$$

Now if inverse is correct
we can check that

$$\therefore a\bar{a} \equiv 1 \pmod{m}$$

$$3 \times 5 \equiv 1 \pmod{7}$$

$$15 \equiv 1 \pmod{7}$$

Hence correct.

Now finally finding x .

Now multiply $\bar{a}$ both side

$$3x \equiv 4 \pmod{7}$$

$$\cancel{3} \times \cancel{5} x \equiv 4 \times 5 \pmod{7}$$

$$x \equiv 20 \pmod{7}$$

$$\boxed{x \equiv 6} \quad \underline{\text{Ans}}$$

~~We can also verify~~ we can also verify the value of x .

$$\text{Put } x = 6$$

~~Multiply 6 both side~~

$$3x \equiv 4 \pmod{7}$$

$$3 \times 6 \equiv 4 \pmod{7}$$

$$\boxed{18 \equiv 4 \pmod{7}}$$

Hence verified

Q10

Sol

Show that 937 is an inverse of 13 modulo 2436

$$\text{G.c.d}(a, m) = (13, 2436)$$

$$a = qd + r$$

$$2436 = (187)(13) + 5$$

$$13 = (2)(5) + 3$$

$$5 = (1)(3) + 2$$

$$3 = (1)(2) + 1$$

$$1 = 3 - 2 \cdot 1$$

$$2 = 5 - 3 \cdot 1$$

$$3 = 13 - 2 \cdot 5$$

$$5 = 2436 - 187 \cdot 13$$

$$1 = 3 - 2 \cdot 1 \Rightarrow 1 = 3 - 1 \cdot (5 - 3 \cdot 1) = 3 \cdot 1 - 5 + 3 \cdot 1$$

$$1 = 3 \cdot 2 - 1 \cdot 5 = 2 \cdot (13 - 2 \cdot 5) - 1 \cdot 5 = 2 \cdot 13 - 4 \cdot 5 - 1 \cdot 5$$

$$1 = 2 \cdot 13 - 5 \cdot 5 = 2 \cdot 13 - 5(2436 - 187 \cdot 13)$$

$$1 = 2 \cdot 13 - 5 \cdot 2436 + 935 \cdot 13$$

$$1 = (937)(13) + (-5)(2436)$$

↓

is an inverse

Hence Proved.

② Solve linear congruences using modular inverse.

① $19x = 4 \pmod{141}$

first we have to find inverse of 19

$a = dq + r$	
$141 = (19)(7) + 8$	$1 = 3 - 2 \cdot 1$
$19 = 8 \cdot 2 + 3$	$2 = 8 - 3 \cdot 2$
$8 = 3 \cdot 2 + 2$	$3 = 19 - 8 \cdot 2$
$3 = 2 \cdot 1 + 1$	$8 = 141 - 19 \cdot 7$

$$1 = 3 - 2 \cdot 1 = 3 - 1 \cdot (8 - 3 \cdot 2) = 3 \cdot 3 - 8 \cdot 1$$

$$1 = 3 \cdot (19 - 8 \cdot 2) - 8 \cdot 1 = 3 \cdot 19 - 6 \cdot 8 - 8 \cdot 1 = 3 \cdot 19 - 7 \cdot 8$$

$$1 = 3 \cdot 19 - 7(141 - 19 \cdot 7) = 3 \cdot 19 - 7 \cdot 141 + 49 \cdot 19$$

$$1 = \underbrace{52 \cdot 19}_{\text{Inverse}} + (-7)(141)$$

Thus $52 \cdot 19 \equiv 1 \pmod{141}$, hence 52 is an inverse of 19.

Now multiply 52 both side.

$$52 \cdot 19x \equiv 52 \cdot 4 \pmod{141}$$

$$x \equiv 208 \pmod{141}$$

$$\boxed{x = 67}$$

It follows that the solution are the integers x satisfying

$$x = (52 \cdot 19x - 7 \cdot 141x) \equiv 208 \equiv 67 \pmod{141}$$

$$x \equiv 2 \pmod{3}; x \equiv 1 \pmod{4}; x \equiv 3 \pmod{5}$$

Given: $a_1 = 2, m_1 = 3; a_2 = 1, m_2 = 4; a_3 = 3, m_3 = 5$

Find: $m = ?; M_1, M_2, M_3 = ?; y_1, y_2, y_3 = ?$

1st we find m ; $m = m_1 \times m_2 \times m_3 = 3 \times 4 \times 5 = 60$

$$\text{Now } M_1 = \frac{m}{m_1} = \frac{60}{3} = 20; M_2 = \frac{m}{m_2} = \frac{60}{4} = 15; M_3 = \frac{60}{5} = 12$$

Now y_1

$$y_1 = 20 \pmod{3}$$

$$a = da + x$$

$$20 = 3 \times 6 + 2 \quad | \quad 1 = 3 - 2 \cdot 1$$

$$3 = 2(1) + 1 \quad | \quad 2 = 20 - 3 \cdot 6$$

$$1 = 3 - 1 \cdot (20 - 3 \cdot 6)$$

$$1 = 3 - 1 \cdot 20 + 3 \cdot 6$$

$$1 = 7 \cdot 3 + (-1)(20)$$

$$\bar{a} + m$$

Hence $-1 + 3 = \boxed{2}$ is an inverse

Now y_3

$$y_3 = 12 \pmod{5}$$

$$a = da + x$$

$$12 = (5)(2) + 2 \quad | \quad 1 = 5 - 2 \cdot 2$$

$$5 = (2)(2) + 1 \quad | \quad 2 = 12 - 5 \cdot 2$$

$$1 = 5 - 2 \cdot (12 - 5 \cdot 2)$$

$$1 = 5 - 2 \cdot 12 + 4 \cdot 5 = (-2)(12) + 5 \cdot 5$$

$$\text{Now } \bar{a} + m$$

$-2 + 5 = 3$ is an inverse

Now y_2

$$y_2 = 15 \pmod{4}$$

$$a = da + x$$

$$15 = 4 \times 3 + 3 \quad | \quad 1 = 4 - 3 \cdot 1$$

$$4 = 3(1) + 1 \quad | \quad 3 = 15 - 4 \cdot 3$$

$$1 = 4 - 1 \cdot (15 - 4 \cdot 3) = 4 - 15 + 4 \cdot 3$$

$$1 = (-1)(15) + 4 \cdot 4$$

$$\bar{a} + m$$

Here $-1 + 4 = \boxed{3}$ is an inverse

$$x = a_1 M_1 y_1 + a_2 M_2 y_2 + a_3 M_3 y_3$$

$$x = (2 \times 20 \times 2) + (1 \times 15 \times 3) + (3 \times 12 \times 3)$$

$$x = 80 + 45 + 108 = 233 \pmod{60}$$

$$\boxed{x = 53}$$

$$x \equiv 2 \pmod{3}, x \equiv 3 \pmod{5}, x \equiv 2 \pmod{7}$$

Given $a_1 = 2, m_1 = 3$; $a_2 = 3, m_2 = 5$; $a_3 = 2, m_3 = 7$
 find $x = ?$ $y_1 = ?$, $y_2 = ?$ & $y_3 = ?$ $M_1, M_2, M_3, m = ?$

$$\text{let } m = m_1 * m_2 * m_3$$

$$m = 3 * 5 * 7 = 105$$

Now,

$$M_1 = \frac{m}{m_1} = \frac{105}{3} = 35$$

$$M_2 = \frac{m}{m_2} = \frac{105}{5} = 21$$

$$M_3 = \frac{m}{m_3} = \frac{105}{7} = 15$$

$$x = a_1 M_1 y_1 + a_2 M_2 y_2 + a_3 M_3 y_3$$

$$x = 2 \cdot 35 \cdot 2 + 3 \cdot 21 \cdot 1 + 2 \cdot 15 \cdot 1 = 233$$

$$x = 233 \pmod{105}$$

$$x = 23$$

Answer

Now,

for y_k we have to find inverse of $M_k \pmod{m_k}$

where $k = 1, 2, 3, \dots$

$$y_k = \overline{M_k} \pmod{m_k}$$

$$y_1 = \overline{35} \pmod{3}$$

↓ Inverse

$$a = qd + r$$

$$35 = (11)(3) + 2$$

$$3 = (1)(2) + 1$$

$$2 = (1)(2) + 0$$

$$\rightarrow 1 = 3 - 2 \cdot 1$$

$$2 = 35 - 11 \cdot 3$$

$$1 = 3 - 1 \cdot (35 - 11 \cdot 3)$$

$$1 = 3 - 1 \cdot 35 + 11 \cdot 3$$

$$1 = -10 \cdot 3 + 11 \cdot 35$$

$$1 = (-1)(35) + (11)(3)$$

Inverse can't be negative

So add m into it

$$-1 + 3 = 2 \text{ is an inverse}$$

$$y_2 = \overline{21} \pmod{5}$$

$$a = qd + r$$

$$21 = (4)(5) + 1$$

$$5 = (1)(5) + 0$$

$$\Rightarrow 1 = (1)(21) + (-4)(5)$$

↓ is an inverse

$$y_3 = \overline{15} \pmod{7}$$

$$a = dq + r$$

$$15 = (2)(7) + 1$$

$$7 = (1)(7) + 0$$

$$\Rightarrow 1 = (1)(15) + (-2)(7)$$

is an inverse

$x = 4 \pmod{5}$ $x = 6 \pmod{8}$ $x = 8 \pmod{9}$	<u>Given:</u> $a_1 = 4, m_1 = 5, a_2 = 6, m_2 = 8$ $a_3 = 8, m_3 = 9$ find: $m = ?$, $M_1 = ?$, $M_2 = ?$, $M_3 = ?$ $y_1 = ?$, $y_2 = ?$, $y_3 = ?$
--	--

1st we find m ; $m = m_1 * m_2 * m_3 = 5 * 8 * 9 = 360$

$$M_1 = \frac{m}{m_1} = \frac{360}{5} = 72; \quad M_2 = \frac{m}{m_2} = \frac{360}{8} = 45; \quad M_3 = \frac{m}{m_3} = \frac{360}{9} = 40$$

Now $y_1, y_2, y_3 = ?$

$$y_k = \overline{M_k} \pmod{m}$$

For $y_1 = ?$

inverse??

$$y_1 = \overline{72} \pmod{5}$$

$$a = qd + r$$

$$72 = (14)(5) + 2$$

$$(5) = (2)(2) + 1$$

$$\Rightarrow 1 = 5 - 2 \cdot 2; \quad 2 = 72 - 14 \cdot 5$$

$$1 = 5 - 2(72 - 14 \cdot 5)$$

$$1 = 5 - 2 \cdot 72 + 28 \cdot 5 = 29(5) + 72(-2)$$

since

-2 is negative, hence have to add $m = 5$

So, $-2 + 5 = 3$ is an inverse

For y_2 :

$$y_2 = \overline{45} \pmod{8}$$

$$a = qd + r$$

$$45 = (5)(8) + 5$$

$$8 = (1)(5) + 3$$

$$5 = (1)(3) + 2$$

$$3 = (1)(2) + 1$$

$$1 = 3 - 2 \cdot 1; \quad 2 = 5 - (1)(3)$$

$$3 = 8 - (1)(5); \quad 5 = 45 - 5 \cdot 8$$

Now

$$1 = 3 - 2(1) = 3 - (1)(5 - 1 \cdot 3)$$

$$1 = 3 - 5 + 1 \cdot 3 = 2 \cdot 3 - 5 \cdot 1$$

$$1 = 2(8 - 5 \cdot 1) - 5 \cdot 1 = 2 \cdot 8 - 5 \cdot 2 - 5 \cdot 1$$

$$1 = 2 \cdot 8 - 5 \cdot 3 = 2 \cdot 8 - 3(45 - 5 \cdot 8)$$

$$1 = 2 \cdot 8 - 3 \cdot 45 + 15 \cdot 8 = 17 \cdot 8 + (-3)(45)$$

Since -3 so we have to add m , Therefore $-3 + 8 = 5$ is an inverse

For y_3 :

$$y_3 = \overline{40} \pmod{9}$$

$$a = qd + r$$

$$40 = (4)(9) + 4$$

$$9 = (2)(4) + 1$$

$$4 = (4)(1) + 0$$

$$4 = 40 - 4 \cdot 9$$

$$1 = 9 - 2 \cdot 4$$

$$1 = 9 - 2(40 - 4 \cdot 9) = 9 - 2 \cdot 40 + 8 \cdot 9$$

$$1 = 9 \cdot 9 - 2 \cdot 40$$

$$\text{inverse} = -2 + 9 = 7$$

Putting values into formula

$$x = 4454 \pmod{360}$$

$$x = 134$$

Fermat's Little Theorem

$$\therefore a^{p-1} \equiv 1 \pmod{p} \quad \therefore p \text{ is Prime}$$

$$\text{Find } 2^{50} \pmod{17}$$

$$\underline{\text{Sol}} \quad 2^{17-1} \equiv 1 \pmod{17} \Rightarrow \boxed{2^{16} \equiv 1 \pmod{17}}$$

$$\Rightarrow 2^{50} \pmod{17} = 2^{16 \times 3 + 2} \pmod{17} \quad \begin{array}{l} a = qd + r \\ 50 = 16 \times 3 + 2 \end{array}$$

$$\equiv (2^{16})^3 \cdot 2^2 \pmod{17}$$

$$\equiv (1)^3 \cdot 4 \pmod{17}$$

$$\equiv 4 \text{ Ans}$$

$$\text{Find } 4^{532} \pmod{11}$$

$$\underline{\text{Sol}} \quad 4^{11-1} \equiv 1 \pmod{11} \Rightarrow \boxed{4^{10} \equiv 1 \pmod{11}}$$

$$\begin{aligned} \Rightarrow 4^{532} \pmod{11} &= 4^{10 \times 53 + 2} \pmod{11} \\ &\equiv (4^{10})^{53} \cdot 4^2 \pmod{11} \\ &\equiv (1)^{53} \cdot 4^2 \pmod{11} \\ &\equiv 16 \pmod{11} \\ &\equiv 5 \text{ Ans} \end{aligned}$$

Find $5^{300} \bmod 7$

Sol:

$$5^{7-1} \equiv 1 \pmod{7} \Rightarrow \boxed{5^6 \equiv 1 \pmod{7}}$$

$$\begin{aligned} 5^{300} \bmod 7 &\equiv 5^{6 \times 50 + 0} \bmod 7 \\ &\equiv (5^6)^{50} \cdot 5^0 \bmod 7 \\ &\equiv (1)^{50} \cdot 1 \bmod 7 \\ &\equiv 1 \bmod 7 \equiv 1 \text{ Ans} \end{aligned}$$

Find $7^{222} \bmod 11$

Sol:

$$7^{11-1} \equiv 1 \pmod{11} \Rightarrow \boxed{7^{10} \equiv 1 \pmod{11}}$$

$$\begin{aligned} 7^{222} \bmod 11 &\equiv 7^{22 \times 10 + 2} \bmod 11 \\ &\equiv (7^{10})^{22} \cdot 7^2 \bmod 11 \\ &\equiv (1)^{22} \cdot 49 \bmod 11 \\ &\equiv 49 \bmod 11 \\ &\equiv 5 \text{ Ans} \end{aligned}$$